

**BITWISE TRIANGULAR COORDINATES FOR
CENTRAL PRODUCTS OF QUATERNION GROUPS
FLORETION BASE VECTORS, DIGITWISE S_3 -ACTIONS, AND
CENTRALIZER TILES**

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This note studies a concrete bitwise and triangular coordinate model for the central product of n copies of the quaternion group Q_8 . The positive basis elements are words of length n in the alphabet $\{1, 2, 4, 7\}$, where the four digits are identified with the quaternionic symbols i, j, k, e . The signed basis group $\mathcal{F}_n$ is therefore not a new abstract family of finite groups, and the real algebra generated by the basis words is the familiar tensor product $\mathcal{A}_n \simeq \mathbb{H}^{\otimes n}$. The contribution is the coordinate model: in this basis, Boolean multiplication, triangular tiling geometry, digitwise S_3 -symmetry, reflection anti-automorphisms, and centralizer structure become visible in the same language.

The main chain of results is as follows. A local rule based on XNOR and AND recovers quaternionic basis multiplication and therefore gives a table-free digitwise multiplication rule in every order. The associated centroid map from basis words to a recursive triangular tiling is equivariant for the digitwise S_3 -action and the dihedral action on the triangle. Odd permutations in this S_3 -action reverse multiplication order; consequently, products of elements symmetric about triangular axes are governed by ordinary or twisted commutation conditions. Synchronized cyclic changes of one or more noncentral digits give equilateral triangles of centroids. The same coordinates also give centralizer tile sets: the centralizer of a

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non-unit basis word has cardinality 4^n in the signed group and occupies exactly one half of the positive tiles. A short parity subsection records the corresponding conjugation-and-cancellation principle for even and odd parts of the tiling.

1. Introduction

The quaternion group $Q_8 = \{\pm i, \pm j, \pm k, \pm e\}$ admits a very compact four-symbol basis notation. In this note we use the alphabet

$$\{1, 2, 4, 7\},$$

identified with the quaternionic symbols by

$$1 \leftrightarrow i, \quad 2 \leftrightarrow j, \quad 4 \leftrightarrow k, \quad 7 \leftrightarrow e.$$

Words of length n in this alphabet are multiplied coordinate by coordinate, with the local quaternionic signs collected into one global sign. We call these words *floreton base vectors* of order n .

The abstract group obtained in this way is standard. It is the central product of n copies of Q_8 , equivalently the extraspecial 2-group specified by that central product. The real algebra generated by the basis words is

$$\mathcal{A}_n \cong \mathbb{H}^{\otimes n},$$

and after complexification one obtains

$$\mathcal{A}_n \otimes_{\mathbb{R}} \mathbb{C} \cong M_{2^n}(\mathbb{C}).$$

Thus the purpose of the paper is not to introduce a new abstract algebra or a new abstract finite group. Rather, it studies a distinguished word basis for $\mathcal{A}_n$ and the structures that become visible in this basis. The same coordinates support a Boolean multiplication rule, a recursive triangular tiling, digitwise S_3 -actions, reflection symmetries, and centralizer tile sets. The payoff is a concrete finite model in which multiplication and symmetry can be studied simultaneously.

The starting point is a two-bit encoding of the four basis symbols. In this encoding the unsigned part of the local product is given by bitwise XNOR, while the sign is determined by a short Boolean expression involving XNOR and AND. Proposition 2.5 identifies this rule with the usual quaternion multiplication table. The Boolean form gives a table-free local multiplication rule that can be applied independently in each coordinate and then assembled by collecting signs. It also supports efficient computation, since multiplication in $\mathcal{A}_n$ can be organized by precomputed index and sign arrays.

The second ingredient is geometric. The same alphabet $\{1, 2, 4, 7\}$ labels the four subtriangles obtained by joining the midpoints of the sides of an equilateral triangle: three corner subtriangles labelled by 1, 2, 4, and a central subtriangle labelled by 7. Iterating this subdivision gives a recursive triangular tiling. The centroid map

$$P : \Delta_n \longrightarrow \mathbb{R}^2$$

associates to each basis word the centroid of its tile. A key point is that the central digit 7 produces no displacement of the centroid but reverses the orientation of all subsequent choices.

The results proved below can be summarized as follows.

1. The Boolean multiplication rule based on XNOR and AND agrees with quaternionic basis multiplication in order one. Since multiplication in order n is digitwise, this gives the rule in every order.
2. The centroid map is equivariant with respect to the digitwise action of S_3 on the digits $\{1, 2, 4\}$ and the dihedral action on the equilateral triangle.
3. Even digit permutations act by algebra automorphisms, while odd digit permutations act by algebra anti-automorphisms. This yields a twisted commutation criterion for products of elements symmetric about possibly different triangular axes; in the same-axis case the criterion reduces to ordinary commutation.
4. The cyclic action $1 \mapsto 2 \mapsto 4 \mapsto 1$, applied globally or to selected coordinates, produces equilateral triangles of centroids.
5. For every non-unit basis word b , the centralizer in the signed group has cardinality 4^n ; among the positive basis words, the corresponding tile set has cardinality $4^n/2$. A short parity subsection in the centralizer section records the related cancellation principle for the even and odd parts of the tiling.

The terminology of floretions arose from earlier computational experiments with the order-two case and with integer sequences obtained from powers or coefficient extractions. Those topics are not needed for the main coordinate model developed here, and are better treated as separate applications or historical notes.

2. Base vectors and multiplication

Definition 2.1 (Base vectors). Fix $n \geq 1$. A floretion base vector of order n is a word

$$b = b_1 b_2 \cdots b_n$$

of length n in the alphabet $\{1, 2, 4, 7\}$. The set of all positive base vectors of order n is denoted by

$$\Delta_n = \{1, 2, 4, 7\}^n.$$

The identity element is the word

$$e_n := 77 \cdots 7.$$

Definition 2.2 (Signed group and real algebra). Set

$$\mathcal{F}_n := \{\pm b : b \in \Delta_n\}.$$

This is the signed group of base vectors of order n . We also write

$$\mathcal{A}_n := \left\{ \sum_{b \in \Delta_n} q_b b : q_b \in \mathbb{R} \right\}$$

for the real algebra generated by Δ_n , with product extended bilinearly from the multiplication of base vectors.

Remark 2.3 (Relation with central products and tensor products). For $n = 1$ the signed basis group is Q_8 . More generally, $\mathcal{F}_n$ is the central product of n copies of Q_8 . Explicitly, it is obtained from Q_8^n by quotienting the central subgroup

$$K_n = \{(\varepsilon_1, \dots, \varepsilon_n) : \varepsilon_r \in \{\pm 1\}, \varepsilon_1 \cdots \varepsilon_n = 1\}.$$

Thus

$$\mathcal{F}_n \simeq Q_8^n / K_n, \quad |\mathcal{F}_n| = \frac{8^n}{2^{n-1}} = 2 \cdot 4^n.$$

The real algebra generated by the basis words is the tensor product

$$\mathcal{A}_n \cong \mathbb{H}^{\otimes n},$$

where a word $b_1 \cdots b_n$ corresponds to the tensor of the associated quaternionic basis symbols. Hence

$$\mathcal{A}_n \otimes_{\mathbb{R}} \mathbb{C} \cong M_{2^n}(\mathbb{C}).$$

These identifications place the construction in standard finite group and associative algebra theory. The digitwise description is retained because it is the one that interacts directly with the Boolean multiplication rule and the triangular tiling.

octal digit	three-bit form	two-bit code	quaternionic symbol
1	001	(00)	i
2	010	(01)	j
4	100	(10)	k
7	111	(11)	e

Table 1: Local encoding of the four allowed digits. The two-bit code is the one used in the XNOR/AND rule.

2.1. The local quaternionic rule

We use the two-bit encoding shown in Table 1. In octal notation this is equivalent to the three-bit representation

$$1 = 001, \quad 2 = 010, \quad 4 = 100, \quad 7 = 111,$$

where the XOR of the three bits is 1.

Definition 2.4 (Bitwise rule in order one). Let

$$x = (ab), \quad y = (cd),$$

with $a, b, c, d \in \{0, 1\}$. Define

$$x \cdot y = (-1)^{m+1} (a \odot c) (b \odot d),$$

where $\odot$ denotes bitwise XNOR and

$$m = (b \& c) + ((a \odot b) \& d) + (a \& (c \odot d)).$$

Here $\&$ denotes bitwise AND.

$\cdot$	i	j	k	e
i	$-e$	k	$-j$	i
j	$-k$	$-e$	i	j
k	j	$-i$	$-e$	k
e	i	j	k	e

Table 2: Quaternionic multiplication in the convention used in this note.

Proposition 2.5 (Local equivalence with quaternions). *The rule of the preceding definition agrees with the quaternionic multiplication table in Table 2.*

Proof. The verification is direct on the sixteen possible products. The unsigned part of the product is

$$(a \odot c)(b \odot d),$$

while the sign is determined by the parity of

$$m = (b \& c) + ((a \odot b) \& d) + (a \& (c \odot d)).$$

For example,

$$k \cdot j = (10)(01).$$

The unsigned part is

$$(1 \odot 0, 0 \odot 1) = (0, 0) = i.$$

The three sign contributions are

$$0 \& 0 = 0, \quad (1 \odot 0) \& 1 = 0, \quad 1 \& (0 \odot 1) = 0.$$

Hence $m = 0$, so the sign is $(-1)^{m+1} = -1$, and $k \cdot j = -i$. The remaining cases give exactly Table 2. $\square$

Definition 2.6 (Digitwise multiplication in order n). Let

$$b = b_1 \cdots b_n, \quad c = c_1 \cdots c_n$$

be two base vectors in Δ_n . For each position r , write the local quaternionic product as

$$b_r c_r = \varepsilon_r d_r, \quad \varepsilon_r \in \{\pm 1\}, \quad d_r \in \{1, 2, 4, 7\}.$$

We define

$$bc = \left(\prod_{r=1}^n \varepsilon_r \right) d_1 d_2 \cdots d_n.$$

Theorem 2.7 (Bitwise rule in every order). *Applying the local bitwise rule in each position and collecting the local signs gives exactly the multiplication of base vectors of order n .*

Proof. By Proposition 2.5, the bitwise rule computes each local product $b_r c_r$ in order one, including its sign. The definition of order- n multiplication is precisely to multiply position by position and collect the product of the local signs. Thus correctness in order one implies correctness in every order. $\square$

Example 2.8. For instance,

$$iji \cdot jek = (i \cdot j)(j \cdot e)(i \cdot k) = k j (-j) = -k j j.$$

3. The centroid algorithm

The basic geometric step is a map from base vectors to points of the plane. The point associated with a word will be the centroid of the corresponding triangular tile.

Definition 3.1 (Elementary vectors). Set

$$\begin{aligned} v(1) &= (\cos 330^\circ, \sin 330^\circ), \\ v(2) &= (\cos 90^\circ, \sin 90^\circ), \\ v(4) &= (\cos 210^\circ, \sin 210^\circ), \end{aligned}$$

and

$$v(7) = (0, 0).$$

Definition 3.2 (Centroid map). Fix an initial scale $d_1 > 0$, and let $b = b_1 \cdots b_n \in \Delta_n$. Define

$$P(b) = \sum_{r=1}^n \sigma_r d_r v(b_r), \quad d_r = \frac{d_1}{2^{r-1}},$$

where $\sigma_r \in \{\pm 1\}$ is the sign accumulated from the digits 7 occurring before position r :

$$\sigma_r = (-1)^{\#\{s < r : b_s = 7\}}.$$

Remark 3.3. The digit 7 produces no displacement of its own, but it reverses the direction of all subsequent displacements. This reversal is essential for compatibility with the recursive triangular tiling.

Definition 3.4 (Orientation of a tile). For $b \in \Delta_n$, set

$$N_{124}(b) := \#\{r : b_r \in \{1, 2, 4\}\}.$$

The tile associated with b is oriented upward if and only if

$$N_{124}(b) \equiv n \pmod{2}.$$

Otherwise it is oriented downward.

Remark 3.5. The map P determines only the centroid. The orientation rule supplies the additional information needed to determine the displayed equilateral triangle.

Proposition 3.6 (Compatibility with the recursive triangular tiling). *Let $T_\emptyset$ be an initial equilateral triangle centered at the origin, and let $R_0 > 0$ be its circum-radius. At each step subdivide an equilateral triangle into the four equilateral*

triangles obtained by joining the midpoints of its sides: three corner triangles, labelled by 1, 2, 4, and the central triangle, labelled by 7. The corner directions are those of the elementary vectors $v(1), v(2), v(4)$.

Put $d_1 = R_0/2$ and $d_r = d_1/2^{r-1}$. If T_b is the triangle in the recursive tiling determined by the word $b = b_1 \cdots b_n \in \Delta_n$, then the centroid of T_b is $P(b)$. Moreover T_b is oriented upward if and only if

$$N_{124}(b) \equiv n \pmod{2}.$$

Proof. We argue by induction on the length of the word. The empty word corresponds to the initial triangle, whose centroid is the origin. Suppose the centroid of the triangle associated with the prefix $b_1 \cdots b_{r-1}$ is known. At level $r-1$, the circumradius of the current triangle is $R_0/2^{r-1}$, and the centroid of a corner subtriangle lies at distance

$$\frac{1}{2} \frac{R_0}{2^{r-1}} = \frac{R_0}{2^r} = d_r$$

from the centroid of the current triangle, in the direction of the corresponding vertex.

At step r , the directions have been reversed once for each earlier occurrence of the central digit 7. Hence the direction sign is

$$\sigma_r = (-1)^{\#\{s < r : b_s = 7\}}.$$

If $b_r \in \{1, 2, 4\}$, the corresponding corner choice adds the displacement $\sigma_r d_r v(b_r)$. If $b_r = 7$, the central subtriangle is chosen: the centroid is unchanged, but the orientation is reversed for subsequent choices. Summing the contributions over $r = 1, \dots, n$ gives the formula for $P(b)$.

Finally, the three corner subtriangles have the same orientation as the current triangle, while the central subtriangle has the opposite orientation. Thus the final orientation is upward if and only if the number of digits 7 is even. Since

$$\#\{r : b_r = 7\} = n - N_{124}(b),$$

this condition is equivalent to $N_{124}(b) \equiv n \pmod{2}$. □

Remark 3.7. The factor σ_r is not needed merely for S_3 -equivariance: the map obtained by putting all $\sigma_r = 1$ would also be S_3 -equivariant. Its role is instead to make the formula compatible with the recursive triangular tiling. The digit 7 selects the central triangle; it produces no displacement of the centroid, but it reverses the orientation for later choices.

Lemma 3.8 (No cancellation of the first nonzero step). *If $b \neq 77 \cdots 7$, then $P(b) \neq 0$.*

Proof. Let m be the first position such that $b_m \neq 7$. The step at position m has length d_m . The sum of the lengths of all subsequent steps is

$$d_{m+1} + \cdots + d_n < d_m.$$

Therefore the first nonzero step cannot be cancelled by the remaining tail of the path, regardless of the later directions. Hence $P(b) \neq 0$. $\square$

4. The digitwise S_3 -action and triangle symmetries

The identification between permutations of the three vertices and the dihedral symmetries of an equilateral triangle is standard; see, for example, [2].

Definition 4.1 (Digitwise action). The group S_3 acts on Δ_n by permuting the digits $\{1, 2, 4\}$ in each position and fixing the digit 7.

Definition 4.2 (Dihedral action). Let D_3 be the symmetry group of the equilateral triangle generated by the three elementary directions $v(1), v(2), v(4)$. We identify D_3 with the group of permutations of these three vectors. If $\pi \in S_3$, let $\rho(\pi) \in D_3$ be the unique linear symmetry of the plane such that

$$\rho(\pi)v(a) = v(\pi(a)) \quad \text{for } a \in \{1, 2, 4\}.$$

Also set $\rho(\pi)v(7) = v(7) = 0$.

Theorem 4.3 (Equivariance of the centroid map). *For every $\pi \in S_3$ and every $b \in \Delta_n$,*

$$P(\pi b) = \rho(\pi)P(b),$$

where πb denotes the digitwise application of π to the word b .

Proof. Write $b = b_1 \cdots b_n$. Since π fixes the digit 7, the positions of the digits 7 in b and in πb are identical. Hence the same coefficients σ_r and the same lengths d_r occur in the formulas for $P(b)$ and $P(\pi b)$. Using the definition of $\rho(\pi)$, we obtain

$$\begin{aligned} P(\pi b) &= \sum_{r=1}^n \sigma_r d_r v(\pi(b_r)) \\ &= \sum_{r=1}^n \sigma_r d_r \rho(\pi)v(b_r) \\ &= \rho(\pi) \sum_{r=1}^n \sigma_r d_r v(b_r) = \rho(\pi)P(b). \end{aligned}$$

$\square$

Corollary 4.4 (Reflections and transpositions). *The digitwise transposition (24) corresponds to reflection in the axis determined by 1; the transposition (14) corresponds to reflection in the axis determined by 2; and the transposition (12) corresponds to reflection in the axis determined by 4.*

Proof. This is the preceding theorem in the special case where π is a transposition. $\square$

Proposition 4.5 (Reflections reverse multiplication order). *We use the same notation for a digitwise permutation of $\{1, 2, 4, 7\}$ and for its linear extension to $\mathcal{A}_n$. If $\pi \in S_3$ is even, then*

$$\pi(XY) = \pi(X)\pi(Y) \quad \text{for all } X, Y \in \mathcal{A}_n.$$

If $\pi \in S_3$ is odd, then

$$\pi(XY) = \pi(Y)\pi(X) \quad \text{for all } X, Y \in \mathcal{A}_n.$$

Proof. It is enough to check the statement on basis words. In order one, the cycle (124), equivalently $i \mapsto j \mapsto k \mapsto i$, preserves the orientation of the quaternionic multiplication table and is an automorphism. For example, $ij = k$ is sent to $jk = i$. On the other hand, a transposition of two of i, j, k reverses the orientation of the table and gives an anti-automorphism. For instance, the transposition fixing i and swapping j and k sends $ij = k$ to $ik = -j$, while reversing the order gives $ki = j$, so the equality is restored in the form $\tau(xy) = \tau(y)\tau(x)$. The remaining products are checked in the same local table. Applying this coordinate by coordinate and collecting the signs gives the assertion for basis words of any order, and bilinearity gives it for all of $\mathcal{A}_n$. $\square$

Proposition 4.6 (Products with different reflection axes). *For $a \in \{1, 2, 4\}$, let τ_a denote the transposition of $\{1, 2, 4\}$ which fixes a and swaps the other two digits; geometrically, τ_a is the reflection in the axis determined by a . Suppose that $X, Y \in \mathcal{A}_n$ satisfy*

$$\tau_a(X) = X, \quad \tau_b(Y) = Y$$

for some $a, b \in \{1, 2, 4\}$. Put

$$\gamma_{ab} := \tau_a \tau_b \in S_3.$$

Then the product XY is symmetric with respect to the a -axis, that is $\tau_a(XY) = XY$, if and only if

$$XY = \gamma_{ab}(Y)X.$$

Similarly, XY is symmetric with respect to the b -axis if and only if

$$XY = Y \gamma_{ba}(X).$$

Proof. Since τ_a is odd, the preceding proposition gives

$$\tau_a(XY) = \tau_a(Y)\tau_a(X) = \tau_a(Y)X.$$

Because Y is fixed by τ_b , we have

$$\tau_a(Y) = \tau_a\tau_b(Y) = \gamma_{ab}(Y).$$

Therefore $\tau_a(XY) = XY$ holds exactly when

$$XY = \gamma_{ab}(Y)X.$$

The second assertion is the same argument with the roles of a and b interchanged:

$$\tau_b(XY) = \tau_b(Y)\tau_b(X) = Y\tau_b(X) = Y\tau_b\tau_a(X) = Y\gamma_{ba}(X).$$

□

Corollary 4.7 (Products of elements symmetric about the same axis). *Let τ be one of the three transpositions in S_3 , and suppose that $X, Y \in \mathcal{A}_n$ are symmetric with respect to the corresponding axis, that is*

$$\tau(X) = X, \quad \tau(Y) = Y.$$

Then

$$\tau(XY) = YX.$$

Consequently XY is symmetric with respect to the same axis if and only if $XY = YX$.

Proof. This is the preceding proposition in the case $a = b$, where $\gamma_{aa} = \tau_a^2 = \text{id}$. □

Remark 4.8. This gives a direct link between reflection symmetry and centralizers. In the same-axis case, preservation of reflection symmetry under multiplication is controlled by the ordinary centralizer. In the different-axis case one obtains a rotated or twisted centralizer condition, such as

$$XY = \gamma_{ab}(Y)X.$$

Thus the geometry of triangular axes detects both ordinary and twisted forms of commutation in the algebra.

The same observation also explains the behavior of inverses. If X is invertible, then any digit permutation or reflection sends X^{-1} to the inverse of the transformed element. In particular, if X is symmetric about a triangular axis and X^{-1} exists, then X^{-1} is symmetric about the same axis. The symmetric units for a fixed axis are therefore closed under inversion, but products remain symmetric only under the commutation condition above.

Example 4.9 (The finite Sierpiński support as a symmetry test). Let

$$S_E(n) := \sum_{b \in \{1,2,4\}^n} b \in \mathcal{A}_n.$$

This is the finite corner-only support obtained by excluding the central digit 7, equivalently by excluding the quaternionic unit e . It is the order- n finite Sierpiński-type support in the triangular tiling [10, 14, 16, 20].

The element $S_E(n)$ is fixed by every digitwise permutation in S_3 , and in particular by every reflection τ . Therefore, if $X \in \mathcal{A}_n$ is symmetric about the corresponding triangular axis, so that $\tau(X) = X$, then the anti-automorphism property gives

$$\tau(XS_E(n)) = \tau(S_E(n))\tau(X) = S_E(n)X.$$

Consequently

$$XS_E(n) \text{ is symmetric about the same axis } \iff XS_E(n) = S_E(n)X.$$

Thus the finite Sierpiński support gives a concrete test case for the preceding centralizer criterion: multiplying a symmetric element by $S_E(n)$ preserves the same reflection symmetry exactly when X centralizes $S_E(n)$.

Corollary 4.10 (The cyclic rotation). *Let $\gamma = (124)$, so that*

$$1 \mapsto 2, \quad 2 \mapsto 4, \quad 4 \mapsto 1, \quad 7 \mapsto 7.$$

Then

$$P(\gamma b) = RP(b),$$

where R is the rotation of the plane sending $v(1)$ to $v(2)$, $v(2)$ to $v(4)$, and $v(4)$ to $v(1)$.

Corollary 4.11 (Global cyclic orbits give equilateral triangles). *Let $b \in \Delta_n$ be a base vector different from the identity $77 \cdots 7$. Then the three points*

$$P(b), \quad P(\gamma b), \quad P(\gamma^2 b)$$

are the vertices of an equilateral triangle.

Proof. By the cyclic-rotation corollary, the three points are

$$P(b), \quad RP(b), \quad R^2P(b),$$

where R is a rotation by 120° or by -120° , depending on the orientation convention. By the no-cancellation lemma, $P(b) \neq 0$. Hence the three points are distinct. Since they are obtained by rotating a nonzero point about the origin through successive angles of 120° , their mutual distances are equal. $\square$

Definition 4.12 (Synchronized local cyclic action). Let $S \subseteq \{1, \dots, n\}$. Define $\gamma_S : \Delta_n \rightarrow \Delta_n$ by applying the cycle $1 \mapsto 2 \mapsto 4 \mapsto 1$ in the coordinates belonging to S , while leaving all other coordinates fixed; the digit 7 is fixed in every coordinate. For $S = \{r\}$ we write γ_r .

Proposition 4.13 (Synchronized local cycles). *Let $S \subseteq \{1, \dots, n\}$, and suppose that b has at least one non-7 digit in a coordinate belonging to S . Then*

$$P(b), \quad P(\gamma_S b), \quad P(\gamma_S^2 b)$$

are the vertices of an equilateral triangle.

Proof. Since γ_S fixes the digit 7, the positions of the digits 7 do not change. Thus the coefficients σ_r in the centroid formula are the same for b , $\gamma_S b$, and $\gamma_S^2 b$. Split the centroid sum into the coordinates outside S and those inside S . Then

$$P(b) = Q + W,$$

where Q is the contribution of the unchanged coordinates and

$$W = \sum_{r \in S} \sigma_r d_r v(b_r).$$

Under γ_S , every nonzero summand of W is rotated by the same 120° rotation R . Hence

$$P(\gamma_S b) = Q + RW, \quad P(\gamma_S^2 b) = Q + R^2 W.$$

It remains only to note that $W \neq 0$. Let m be the first coordinate in S for which $b_m \neq 7$. The summand at m has length d_m , while the sum of the lengths of all later selected summands is strictly smaller than d_m . Therefore the first nonzero selected summand cannot be cancelled by the later selected summands. The three displayed points are therefore a translation of the orbit of a nonzero vector under a 120° rotation, and so form an equilateral triangle. $\square$

Remark 4.14. The global cyclic-orbit statement is the special case $S = \{1, \dots, n\}$. The case $S = \{r\}$ says that a single noncentral digit already determines a small equilateral triangle of possible centroid positions. Thus the equilateral-triangle phenomenon is local as well as global.

5. Centralizer tiles

In this section we distinguish carefully between the centralizer in the signed group $\mathcal{F}_n$ and the corresponding set of positive tiles in Δ_n . The preceding sections show why centralizers are geometrically natural: for two elements symmetric

about the same triangular axis, commutation is exactly the condition that their product preserve that reflection symmetry. The parity subsection below records a related algebraic form of the same cancellation phenomenon before we count the corresponding tile sets.

5.1. Parity, conjugation, and cancellation

Before counting centralizer tiles, we record a parity projection identity that expresses the same sign-cancellation principle algebraically. The orientation of a tile is governed by the parity of the number of noncentral digits, but the following definition separates this parity from the visual up/down convention.

Definition 5.1 (Even and odd parts). For a basis word $b \in \Delta_n$, let

$$v(b) := N_{124}(b) \pmod{2}.$$

For $X = \sum_{b \in \Delta_n} q_b b \in \mathcal{A}_n$, define

$$X_{\text{even}} := \sum_{v(b)=0} q_b b, \quad X_{\text{odd}} := \sum_{v(b)=1} q_b b.$$

Thus $X = X_{\text{even}} + X_{\text{odd}}$. We write $\mathcal{A}_n^{\text{even}}$ and $\mathcal{A}_n^{\text{odd}}$ for the corresponding linear subspaces. When n is even, the even part is the upward part of the tiling and the odd part is the downward part. When n is odd, the two visual orientations are interchanged.

Proposition 5.2 (Parity projection identities). *Let $X = U + D$, where $U = X_{\text{even}}$ and $D = X_{\text{odd}}$. Then:*

$$D = X \implies X^{2m} \in \mathcal{A}_n^{\text{even}} \text{ and } X^{2m+1} \in \mathcal{A}_n^{\text{odd}} \quad (m \geq 0),$$

and

$$U = X \implies X^m \in \mathcal{A}_n^{\text{even}} \quad (m \geq 0).$$

Moreover,

$$(X^2)_{\text{odd}} = 2(X X_{\text{odd}})_{\text{odd}}.$$

Thus the odd part of a square is determined entirely by the interaction between X and its odd component; the even-even and odd-odd contributions do not survive the odd projection.

Proof. Let $\kappa : \mathcal{A}_n \rightarrow \mathcal{A}_n$ be the linear map defined on basis words by

$$\kappa(b) = (-1)^{N_{124}(b)} b.$$

Equivalently, κ applies quaternionic conjugation in each digit. Hence κ is an anti-automorphism:

$$\kappa(XY) = \kappa(Y)\kappa(X).$$

The even and odd parts are the $+1$ and -1 eigenspaces of κ .

If $X = D$, then $\kappa(X) = -X$, and for powers of a single element the reversal of order has no effect:

$$\kappa(X^m) = \kappa(X)^m = (-1)^m X^m.$$

This proves the first assertion. If $X = U$, the same argument gives $\kappa(X^m) = X^m$ for all m , proving the second assertion.

For the last identity, write $X = U + D$. Since $\kappa(U) = U$ and $\kappa(D) = -D$, the anti-automorphism property gives

$$\kappa(X^2) = \kappa(X)\kappa(X) = (U - D)^2.$$

Therefore

$$\begin{aligned} (X^2)_{\text{odd}} &= \frac{X^2 - \kappa(X^2)}{2} \\ &= \frac{(U + D)^2 - (U - D)^2}{2} \\ &= UD + DU. \end{aligned}$$

Similarly,

$$XD = UD + D^2,$$

while

$$\kappa(XD) = \kappa(D)\kappa(X) = (-D)(U - D) = -DU + D^2.$$

Hence

$$(XD)_{\text{odd}} = \frac{XD - \kappa(XD)}{2} = \frac{UD + DU}{2}.$$

Thus $(X^2)_{\text{odd}} = 2(XD)_{\text{odd}}$, as claimed. $\square$

Remark 5.3. For even order this last identity says simply

$$(X^2)_{\text{down}} = 2(X X_{\text{down}})_{\text{down}}.$$

It is another form of the same sign-cancellation principle used in the centralizer counts below.

Definition 5.4 (Centralizer in the signed group). For $g \in \mathcal{F}_n$, set

$$C_{\mathcal{F}_n}(g) := \{h \in \mathcal{F}_n : hg = gh\}.$$

Lemma 5.5 (Conjugacy classes). *If $b \in \Delta_n$ and $b \neq e_n$, then the conjugacy class of b in $\mathcal{F}_n$ is*

$$\{b, -b\}.$$

Proof. In each coordinate, conjugation by a quaternionic basis element sends the local symbol i, j, k, e either to itself or to its negative. Hence every conjugate of b is necessarily b or $-b$.

It remains to see that both occur. Since $b \neq e_n$, there is a position r with $b_r \in \{1, 2, 4\}$. Choose a base vector $c \in \Delta_n$ whose entries are 7 in all positions except position r , where we choose a quaternionic symbol that anticommutes with b_r . Conjugation by c changes exactly one local sign and therefore produces the global element $-b$. $\square$

Theorem 5.6 (Cardinality of the centralizer). *If $b \in \Delta_n$ and $b \neq e_n$, then*

$$|C_{\mathcal{F}_n}(b)| = 4^n.$$

Proof. The group $\mathcal{F}_n$ has order

$$|\mathcal{F}_n| = 2 \cdot 4^n.$$

By the preceding lemma, the conjugacy orbit of b has cardinality 2. The orbit-stabilizer theorem gives

$$|\mathcal{F}_n| = |\text{Orb}(b)| |\text{Stab}(b)|.$$

The stabilizer of the conjugation action is precisely the centralizer $C_{\mathcal{F}_n}(b)$. Therefore

$$2 \cdot 4^n = 2 |C_{\mathcal{F}_n}(b)|,$$

and hence $|C_{\mathcal{F}_n}(b)| = 4^n$. $\square$

Definition 5.7 (Positive centralizer tiles). For $b \in \Delta_n$, define the positive tile set of the centralizer by

$$C_{\text{tiles}}(b) := \{c \in \Delta_n : cb = bc\}.$$

We also decompose $C_{\text{tiles}}(b)$ according to the sign of the common product:

$$C_+(b) := \{c \in \Delta_n : cb = bc \text{ and } cb \in \Delta_n\},$$

$$C_-(b) := \{c \in \Delta_n : cb = bc \text{ and } cb \in -\Delta_n\}.$$

Thus

$$C_{\text{tiles}}(b) = C_+(b) \sqcup C_-(b).$$

Remark 5.8. The symbols $+$ and $-$ in C_+ and C_- refer to the sign of the common product $cb = bc$, not to the distinction between commutation and anticommutation. The set $C_{\text{tiles}}(b)$ is the centralizer as seen on positive tiles only.

Corollary 5.9 (One half of the positive tiling). *If $b \in \Delta_n$ and $b \neq e_n$, then*

$$|C_{\text{tiles}}(b)| = \frac{4^n}{2}.$$

In particular, the centralizer displayed as a set of positive tiles occupies exactly one half of the order- n tiles.

Proof. Each $c \in C_{\text{tiles}}(b)$ contributes two elements to the signed centralizer, namely c and $-c$. Conversely, every element of $C_{\mathcal{F}_n}(b)$ is of the form c or $-c$ with $c \in C_{\text{tiles}}(b)$. Hence

$$|C_{\mathcal{F}_n}(b)| = 2|C_{\text{tiles}}(b)|.$$

The preceding theorem gives the result. $\square$

Lemma 5.10 (Unsigned part of a product). *For all $b, c \in \Delta_n$, the products bc and cb have the same underlying base vector, possibly with different signs.*

Proof. In order one, the unsigned part of the bitwise rule is

$$(a \odot c)(b \odot d),$$

which is symmetric in the two factors because XNOR is commutative. Applying this in each position gives the claim in every order. $\square$

Corollary 5.11 (Sign test). *Let $b, c \in \Delta_n$. Then c centralizes b if and only if bc and cb have the same sign. If the signs are opposite, then $bc = -cb$.*

Proof. By the preceding lemma, the two products have the same underlying base vector. Thus they can differ only by sign. They are equal exactly when the signs agree, and they are opposite exactly when the signs differ. $\square$

Remark 5.12 (Squares and cancellations). Let

$$X = \sum_{c \in \Delta_n} q_c c \in \mathcal{A}_n.$$

Then

$$X^2 = \sum_{c \in \Delta_n} q_c^2 c^2 + \sum_{\{c, d\} \subset \Delta_n, c \neq d} q_c q_d (cd + dc).$$

By the sign test, for two base vectors c, d , the products cd and dc have the same underlying base vector and may differ only by sign. If $cd = -dc$, the mixed contribution $q_c q_d (cd + dc)$ vanishes; if $cd = dc$, it contributes $2q_c q_d cd$. This gives a natural computational form for the square of a floretion and connects the calculation of X^2 directly to the centralizer tile sets described above.

Definition 5.13 (Sums of centralizer components). For $b \in \Delta_n$, set

$$\Sigma_+(b) := \sum_{c \in C_+(b)} c, \quad \Sigma_-(b) := \sum_{c \in C_-(b)} c, \quad \Sigma_{\text{tiles}}(b) := \sum_{c \in C_{\text{tiles}}(b)} c.$$

These sums are regarded as elements of the real algebra $\mathcal{A}_n$.

Proposition 5.14 (Vanishing of the two signed components). *Let $b \in \Delta_n$ satisfy*

$$b^2 = e_n.$$

Then

$$\Sigma_-(b)\Sigma_+(b) = 0 \quad \text{and} \quad \Sigma_+(b)\Sigma_-(b) = 0.$$

Proof. Write $C^\varepsilon(b)$, with $\varepsilon \in \{+1, -1\}$, for $C_+(b)$ or $C_-(b)$, respectively, and set

$$\Sigma_\varepsilon(b) := \sum_{c \in C^\varepsilon(b)} c.$$

If $c \in C^\varepsilon(b)$, then c centralizes b and

$$cb = bc = \varepsilon d$$

for a unique $d \in \Delta_n$. Since $b^2 = e_n$, we obtain

$$db = \varepsilon cb^2 = \varepsilon c.$$

Moreover d still centralizes b , since $d = \varepsilon cb$, and hence

$$bd = \varepsilon bcb = \varepsilon cbb = \varepsilon c,$$

while $db = \varepsilon c$ as above. Therefore $d \in C^\varepsilon(b)$. Thus right multiplication by b , after forgetting the sign, permutes $C^\varepsilon(b)$. Consequently

$$\Sigma_\varepsilon(b)b = \varepsilon \Sigma_\varepsilon(b).$$

Since all elements in $C^\varepsilon(b)$ centralize b , we also have

$$b\Sigma_\varepsilon(b) = \varepsilon \Sigma_\varepsilon(b).$$

Now put

$$X = \Sigma_-(b)\Sigma_+(b).$$

On one hand,

$$Xb = \Sigma_-(b)(\Sigma_+(b)b) = \Sigma_-(b)\Sigma_+(b) = X.$$

On the other hand,

$$bX = (b\Sigma_-(b))\Sigma_+(b) = -\Sigma_-(b)\Sigma_+(b) = -X.$$

But X centralizes b , because both factors lie in the subalgebra generated by elements that centralize b . Hence $Xb = bX$, and the displayed equalities give $X = -X$. Over $\mathbb{R}$, this implies $X = 0$. The proof of $\Sigma_+(b)\Sigma_-(b) = 0$ is identical, with the order of the two factors reversed. $\square$

Example 5.15. For $b = ii$ in order two,

$$C_-(ii) = \{ie, jk, kj, ei\}, \quad C_+(ii) = \{ii, jj, kk, ee\}.$$

Thus

$$\Sigma_-(ii) = ie + jk + kj + ei, \quad \Sigma_+(ii) = ii + jj + kk + ee.$$

Since $(ii)^2 = e_2$, the preceding proposition gives

$$(ie + jk + kj + ei)(ii + jj + kk + ee) = 0.$$

Figure 1 shows a small decomposition of $C_{\text{tiles}}(b)$ into its two signed components. Figure 2 shows the two components for a higher-order example. These pictures are not needed for the proofs, but they make the geometric structure of the centralizer tiles visible.

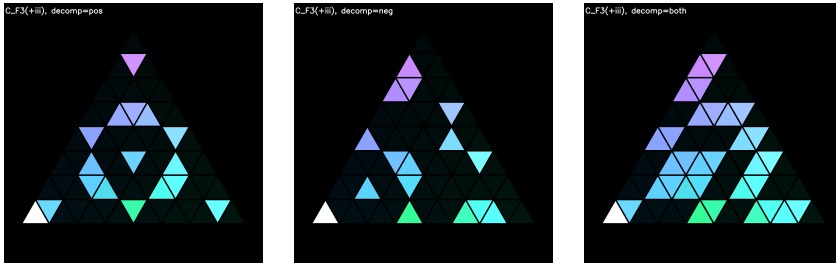


Figure 1: A decomposition of $C_{\text{tiles}}(b)$ into the components $C_+(b)$ and $C_-(b)$ with base vector $b = iii$.

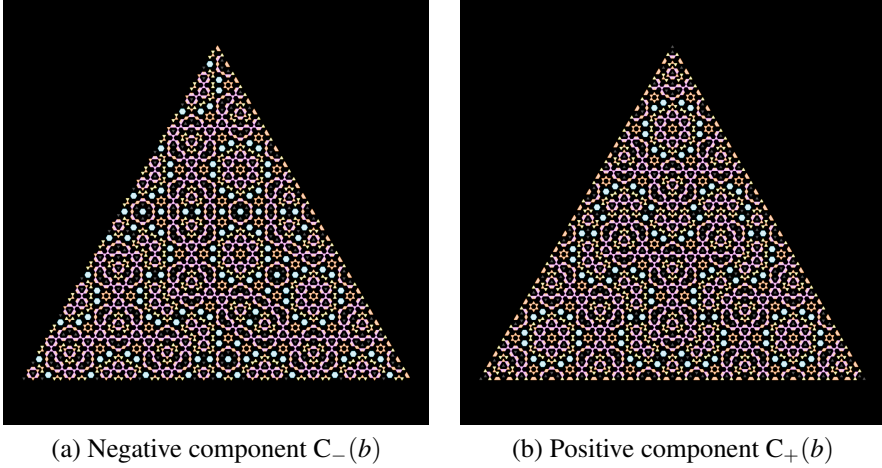


Figure 2: The two centralizer components for the base vector $b = ieiiii$. The visualizations were generated with the Floretion add-on for Blender.

6. Concluding remarks

The guiding point of the note is that the floretion basis is a distinguished coordinate system for the familiar algebra $\mathbb{H}^{\otimes n}$. In this basis, quaternionic multiplication, recursive triangular tilings, digitwise S_3 -actions, reflection symmetries, parity cancellation, and centralizer tile sets can be read in the same notation. The results above should therefore be understood as properties of this coordinate model rather than as claims about a new abstract group or algebra.

The equivariance theorem identifies the symbolic S_3 -action with the geometric action on the equilateral triangle. Since odd digit permutations are algebra anti-automorphisms, triangular reflection symmetry also detects commutation. If two elements are symmetric about the same axis, their product remains symmetric about that axis exactly when the two elements commute. If the axes are different, the corresponding condition is a rotated centralizer relation of the form $XY = \gamma(Y)X$. Thus the triangular geometry does not merely display the basis; it records when products preserve, reverse, or twist reflection symmetry.

The centralizer theorem gives a second manifestation of the same sign structure. For every non-unit basis word, the signed centralizer has cardinality 4^n , and the corresponding positive tile set occupies exactly one half of the order- n tiling. The parity identity

$$(X^2)_{\text{odd}} = 2(X X_{\text{odd}})_{\text{odd}}$$

is a compatible projection form of this cancellation principle. From a computational point of view, the same signs allow products to be organized as signed scalar products using precomputed index and sign arrays.

A natural continuation is to study geometrically defined supports in $\mathcal{A}_n$ and to unfold the triangular picture into the plane. The initial equilateral triangle can be regarded as a fundamental chamber; reflecting it across its three sides begins to generate the full triangular reflection tiling. For the centralizer components $C_+(b)$ and $C_-(b)$, this raises concrete questions: whether side reflections correspond to explicit involutions on the words in $\{1, 2, 4, 7\}^n$, whether $C_+(b)$ and $C_-(b)$ are exchanged or preserved by such involutions, and for which b the unfolded centralizer patterns become periodic under translations of the triangular lattice.

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